

Resistor-Free Miller Compensation of a Class-AB Fully Differential DDA by Feedback to the Cascode Sources

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Abstract—The fully differential difference amplifier (DDA) under review uses a folded-cascode first stage and a class-AB common-source output stage in which every output transistor carries its own Miller capacitor with a series nulling resistor. We ask whether the resistors can be removed by returning each Miller capacitor to the *source* of a cascode transistor instead of the gate of the output device (the cascode drain). The answer is yes, with two conditions. First, the capacitor must land on a cascode source that does not also receive the input-pair signal current; landing on a folding node replaces the right-half-plane (RHP) zero by a symmetric zero pair and degrades gain margin to about 5 dB in our example. Second, the non-dominant poles become a complex pair whose quality factor grows as $\sqrt{g_{m2}}$, so the design must be checked at maximum output current, where a class-AB stage has its largest g_{m2} . We give closed-form design equations and a numerical comparison of the variants.

Index Terms—Frequency compensation, Miller compensation, cascode compensation, current-buffer compensation, indirect feedback compensation, class-AB output stage, differential difference amplifier.

I. QUESTION

The amplifier is a fully differential, folded-cascode DDA [7] (FCDDA) with class-AB outputs. Each output transistor is compensated by a series RC network from its drain (the output) to its gate. The resistor R_z exists only to move the RHP zero at g_{m2}/C_c , which the feedforward current through C_c creates [3]. The question raised in the review was: *Can the resistors be avoided if the feedback is taken from the sources of the cascodes rather than the drains?*

This is cascode, current-buffer, or indirect compensation, first proposed by Ahuja [1] and analysed further in [2], [4], [5], [6]. The point of the technique is that the capacitor current reaches the high-impedance gate node only through the common-gate cascode, which acts as a unidirectional current buffer. The direct feedforward path from the output through C_c into the gate node, which causes the RHP zero, is therefore broken.

II. SMALL-SIGNAL MODEL

Fig. 1 shows the model used for all variants. Node X is the high-impedance gate of one output transistor, with conductance G_1 and capacitance C_1 (mostly the output device's

C_{gs}). The output node has G_2 and load C_L . The cascode is modelled as an ideal current buffer: input conductance g_{mc} and parasitic capacitance C_s at its source S , and an output current $g_{mc}v_S$ into X . Cascode output resistance is neglected; it only adds terms of order $1/(g_{mc}r_o)$.

The first-stage transconductance g_{m1} injects its signal current either at X (cases a, b, d) or at S (case c). Case (c) applies when S is a folding node, where the input pair's drain current is summed with the cascode branch current.

III. ANALYSIS

A. Conventional Miller Compensation (a, b)

With C_c from the output to X ,

$$L_a(s) = \frac{-g_{m1}(g_{m2} - sC_c)}{G_1G_2 + s b_1 + s^2 b_2}, \quad (1)$$

where $b_1 = g_{m2}C_c + (G_1 + G_2)C_c + G_1C_L + G_2C_1$ and $b_2 = C_1C_L + C_c(C_1 + C_L)$. The RHP zero is at $+g_{m2}/C_c$ and the non-dominant pole is at $g_{m2}C_c/b_2 \approx g_{m2}/C_L$. A series resistor moves the zero to $1/[C_c(1/g_{m2} - R_z)]$ and adds a third pole near $1/(R_zC_1)$. For a class-AB stage g_{m2} varies with output current over a wide range, so no fixed R_z cancels the zero at all operating points.

B. Cascode Compensation, Non-Input Cascode Source (d)

Solving the network of Fig. 1 with the input at X gives

$$L_d(s) = \frac{-g_{m1}g_{m2}[g_{mc} + s(C_c + C_s)]}{a_0 + a_1s + a_2s^2 + a_3s^3} \quad (2)$$

with

$$\begin{aligned} a_0 &= G_1G_2g_{mc}, \\ a_1 &= g_{m2}g_{mc}C_c + g_{mc}(C_1G_2 + C_LG_1 + C_cG_1) \\ &\quad + G_1G_2(C_c + C_s), \\ a_2 &= g_{mc}C_1(C_L + C_c) + C_1G_2(C_c + C_s) \\ &\quad + G_1(C_LC_c + C_LC_s + C_cC_s), \\ a_3 &= C_1(C_LC_c + C_LC_s + C_cC_s). \end{aligned}$$

The only zero is in the left half plane:

$$z_d = -\frac{g_{mc}}{C_c + C_s}. \quad (3)$$

No nulling resistor is needed. The cascode's $1/g_{mc}$ acts as a unidirectional nulling resistor.

Technical note, 10 October 2026, prepared for the Chipalooza #3 schematic design review (FCDDA, GF180MCU). Drafted with the assistance of Claude (Anthropic). All transfer functions were derived symbolically (SymPy) and all numbers in Section IV were computed from them; the scripts accompany this note.

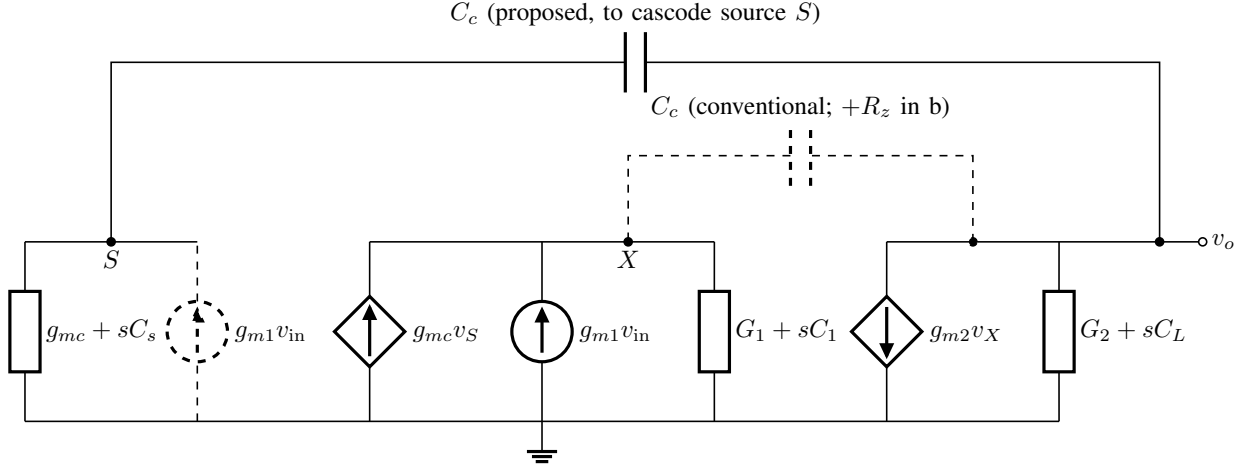


Fig. 1. Small-signal model of one compensated output transistor. The cascode is an ideal current buffer with input admittance $g_{mc} + sC_s$ at its source S . Solid C_c : compensation to the cascode source S (cases c, d). Dashed C_c : conventional Miller compensation to the gate node X (case a; case b adds R_z in series). The first-stage signal current enters at X (solid source; cases a, b, d) or, when S is a folding node, at S (dashed source; case c).

For $G_1, G_2 \rightarrow 0$ and $C_s \rightarrow 0$, the dominant pole and gain-bandwidth product are the same as for Miller compensation:

$$p_1 \approx \frac{G_1 G_2}{g_{m2} C_c}, \quad \omega_u \approx \frac{g_{m1}}{C_c}. \quad (4)$$

The two non-dominant poles are the roots of $C_1 C_L C_c s^2 + g_{mc} C_1 (C_L + C_c) s + g_{m2} g_{mc} C_c = 0$, i.e. a pair with

$$\omega_0 = \sqrt{\frac{g_{m2} g_{mc}}{C_1 C_L}}, \quad (5)$$

$$Q = \frac{C_c}{C_L + C_c} \sqrt{\frac{g_{m2} C_L}{g_{mc} C_1}}. \quad (6)$$

Unlike Miller compensation, pole splitting now places the non-dominant poles at the geometric mean of g_{m2}/C_L and g_{mc}/C_1 , well above g_{m2}/C_L when C_1 is small. The price is that these poles can be complex. In the numerical example, (6) underestimates the exact Q by about 10% because C_s is neglected.

Damping condition. Requiring $Q \leq Q_{\max}$ gives

$$g_{mc} \geq \frac{g_{m2}}{Q_{\max}^2} \frac{C_L}{C_1} \left(\frac{C_c}{C_L + C_c} \right)^2. \quad (7)$$

Since $Q \propto \sqrt{g_{m2}}$, the worst case for a class-AB stage is maximum output current, not the quiescent point. A large output device helps here, because its gate capacitance C_1 lowers Q .

C. Cascode Compensation at a Folding Node (c)

If S also receives the first-stage signal current, the denominator is unchanged but the numerator becomes

$$N_c(s) = g_{m1} (C_1 C_c s^2 + C_c G_1 s - g_{m2} g_{mc}), \quad (8)$$

with zeros at approximately

$$z_c \approx \pm \sqrt{\frac{g_{m2} g_{mc}}{C_1 C_c}}. \quad (9)$$

The input current now splits at S : part goes through the cascode (inverting path through g_{m2}), and part goes through

TABLE I
LOOP AND CLOSED-LOOP ($\beta = 1$) FIGURES OF MERIT

Variant	f_u MHz	PM deg	GM dB	Q	Peak dB	t_s ns
<i>Quiescent, $g_{m2} = 300 \mu\text{S}$</i>						
a Miller, no R_z	12.0	36.0	10.0	–	4.4	228
b Miller + R_z	12.0	48.2	20.9	–	1.8	158
c Casc., folding node	18.7	50.8	5.1	0.89	4.7	155
d Casc., non-input source	21.3	81.3	∞	0.89	0.0	87
<i>Heavy drive, $g_{m2} = 3 \text{ mS}$</i>						
a Miller, no R_z	15.7	81.6	29.6	–	0.0	59
b Miller + R_z	16.7	100.4	∞	0.90	0.0	81
d Casc., non-input source	20.4	122.5 [†]	∞	2.80	5.0	100
d as above, $g_{mc} = 600 \mu\text{S}$	16.3	98.1	∞	1.64	0.0	78

t_s : settling to 0.1% after a step. [†]The loop gain crosses 0 dB three times (20, 73 and 132 MHz), so the phase margin at the first crossing is misleading; the closed-loop response peaks by 5 dB at 137 MHz.

C_c directly into the output (non-inverting feedforward). The near-symmetric zero pair contributes almost no phase, but it raises $|L|$ above $|z_c|$. Together with the complex pole pair this costs gain margin. *The capacitor should therefore not land on a node that carries input signal current.*

IV. NUMERICAL COMPARISON

The parameters are illustrative, chosen to resemble a low-quiescent-current class-AB stage. They are not extracted from the GF180MCU schematic: $g_{m1} = 100 \mu\text{S}$, $g_{m2} = 300 \mu\text{S}$ (quiescent) or 3 mS (heavy drive), $g_{mc} = 200 \mu\text{S}$, $G_1 = 0.1 \mu\text{S}$, $G_2 = 10 \mu\text{S}$, $C_1 = 500 \text{ fF}$, $C_s = 150 \text{ fF}$, $C_L = 2 \text{ pF}$, $C_c = 1 \text{ pF}$, $R_z = 1/g_{m2}|_{\text{quiescent}} = 3.33 \text{ k}\Omega$. All cases use the same C_c , so they have the same nominal ω_u . Closed-loop figures are for unity feedback ($\beta = 1$), the worst case for stability.

Table I and Figs. 2 and 3 show the results.

- At the quiescent point, case (d) is clearly the best: it needs no resistor, has the highest phase margin, and settles fastest. The LHP zero (3), here at 28 MHz, also raises f_u above $g_{m1}/(2\pi C_c)$.

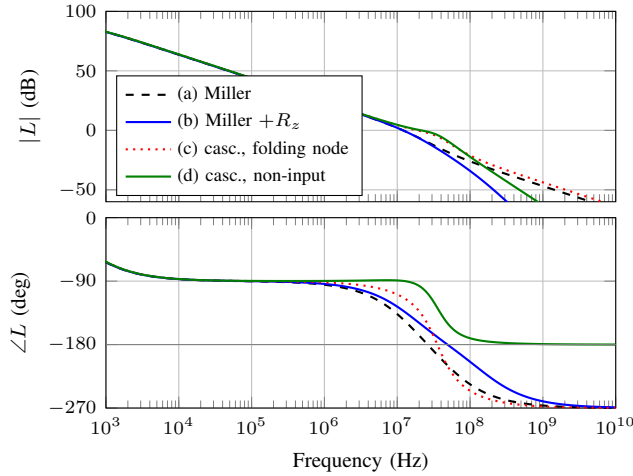


Fig. 2. Loop gain at the quiescent point ($g_{m2} = 300 \mu\text{S}$) for the four compensation variants.

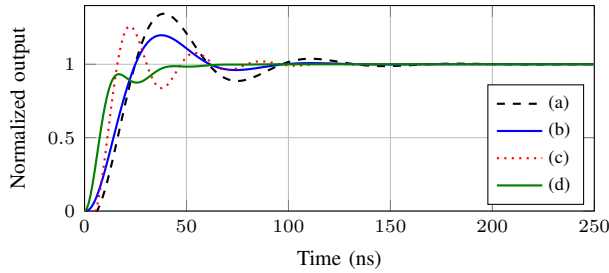


Fig. 3. Small-signal step response in unity feedback at the quiescent point.

- Case (c) is worse than the conventional resistor version in gain margin and overshoot.
- Under heavy drive, case (d) with the quiescent-point g_{mc} develops a $Q \approx 2.8$ resonance that pushes the loop gain above 0 dB again. Tripling g_{mc} to $600 \mu\text{S}$ lowers Q to 1.6 and removes the closed-loop peaking. Reaching $Q_{\max} = 0.7$ would require $g_{mc} \approx 2.7 \text{ mS}$ by (7), or a larger C_1 .

V. APPLICATION TO THE FCDDA

A. Choice of Landing Node

In a folded cascode, each signal branch has two cascode sources. One is the folding node, which also receives the input-pair drain current. The other is the source of the complementary cascode, fed only by a bias current source. Only the second is a valid landing node for C_c .

In the schematic shown at the review, the nets `sum_p` and `sum_n`, where the two input pairs' currents are combined, appear to be folding nodes. If so, they are *not* suitable. This assessment is tentative; it is based on a screenshot and should be confirmed against the netlist.

B. Class-AB Specifics

Each output transistor (PMOS and NMOS) needs a path from the output to its gate. With a floating class-AB control (the NSUP/PSUP devices between the cascode drains), the

signal current injected at one cascode source reaches both gates. Two options should be compared in simulation:

- 1) Return both capacitors to the non-input cascode sources, relying on the floating class-AB control to couple the gates.
- 2) Return one capacitor to the non-input cascode source and the other to the folding node, and check for the zero-pair degradation of case (c).

Equation (7) should be evaluated with the largest g_{m2} that occurs, i.e. at maximum output current. The effective C_1 is the gate capacitance of the device that conducts.

C. Other Effects

- *Pole at S.* The cascode source must stay a low-impedance node: g_{mc} must be large compared with $\omega_0(C_s + C_c)$, which favours wide cascodes and adequate branch current.
- *Slew rate.* During slewing the capacitor current comes from the cascode branch. If the branch bias current I_B is smaller than the tail current, the cascode cuts off; the slew rate is then about $\min(I_{\text{tail}}, I_B)/C_c$.
- *Common-mode loop.* The CMFB loop sees the same modified output stage and must be checked separately.
- *PSRR.* Supply coupling through C_c into the gate node is reduced, which was Ahuja's original motivation [1], [2].

VI. RECOMMENDED VERIFICATION

- 1) Run a loop-stability analysis (`stb`) for both the differential-mode and common-mode loops at the quiescent point and at maximum sourcing and sinking output current, across corners.
- 2) Inspect the full $|L(f)|$ for multiple 0 dB crossings; the phase margin at the first crossing is not sufficient (Table I).
- 3) Check large-signal step settling for slew-induced cascode cutoff.
- 4) Size g_{mc} with (7) at maximum g_{m2} , using $Q_{\max} \approx 0.7$.

VII. CONCLUSION

Yes: the nulling resistors can be removed by returning each Miller capacitor to a cascode source. The RHP zero disappears, and an LHP zero at $-g_{mc}/(C_c + C_s)$ takes its place. Two conditions apply. The landing node must not carry input signal current, and g_{mc} must be large enough to damp the resulting complex pole pair at the highest output current. Because the class-AB output g_{m2} varies strongly with load current, eliminating the fixed resistor is attractive here: unlike R_z , the cascode-compensation zero does not depend on g_{m2} .

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